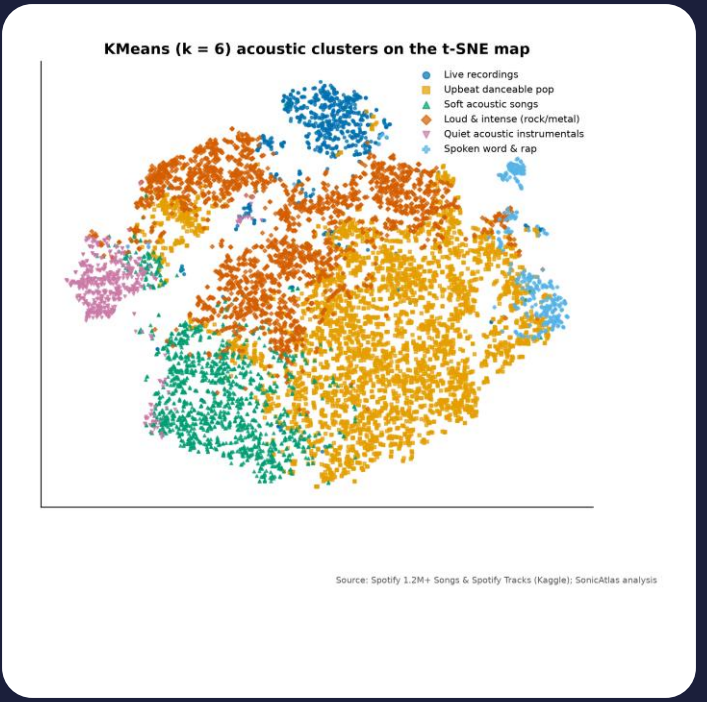
SONICATLAS

An Interactive In-Browser EDA Platform for Spotify Audio Features and Lyrics Across Genres and Eras

Review-2 | Design complete, implementation in progress

Team 05
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Real acoustic map of 10,000 tracks (t-SNE)

BACSE301 Exploratory Data Analysis (Project-based Course)
Faculty: _____
School of Computer Science and Engineering, VIT Vellore | October 2026

We map how the sound of music changes across genres and eras

Abstract

Public Spotify tables describe each track with audio features such as energy, valence and danceability. We clean 1,204,025 tracks (1900-2020) plus a genre table of 89,740 tracks in 114 genres and Genius lyrics, explore them with the course methods, and publish the results in a serverless, in-browser query tool.

Objectives

1. Clean 1M+ tracks; validate MICE by masking 5% of values
2. Descriptive stats in course conventions (n-1, Tukey IQR, G1, G2)
3. Correlation and multivariate genre fingerprints
4. Compare PCA, t-SNE, LLE, UMAP; name acoustic clusters
5. Trends, STL seasonality, change points, forecast
6. ANOVA, chi-square, feature selection; lyrics sentiment
7. In-browser SQL with lite-data queries under 1 second

Prior work is static and studies sound and words apart

Study	What they did	Gap SonicAtlas addresses
J. Serrà et al. (2012)	Tracked pitch, timbre and loudness of Western pop over 50+ years	Static figures; no genre-level or interactive view
M. Interiano et al. (2018)	Linked audio features of 500k UK releases to chart success	No lyrics; results cannot be re-sliced by readers
C. O. Brand et al. (2019)	Measured rising negativity in song lyrics over 50 years	Lyrics only; audio mood not compared
L. van der Maaten et al. (2008)	Introduced t-SNE for 2-D maps of high-dimensional data	Global distances unreliable; no music benchmark
D. H. da Silva et al. (2025)	Streamlit web app for EDA and ML on medical data	Needs a server; not music-aware

Static results

fixed figures, no re-slicing

Sound vs words

audio and lyrics never compared

Data quality

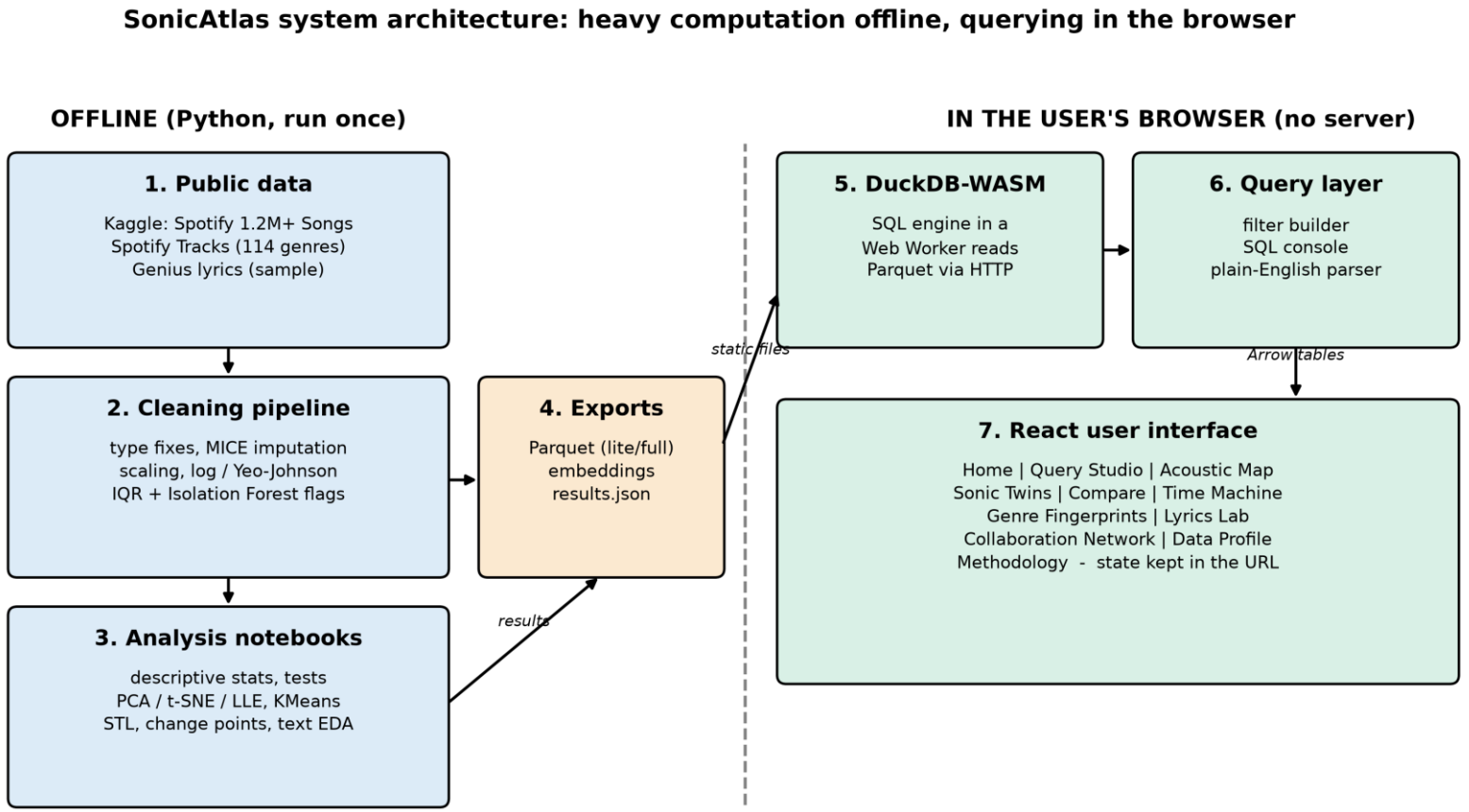
zero tempos, re-issues ignored

Generic tools

not music-aware, need a server

25 verified papers reviewed in the report (Table I).

Heavy maths runs offline; querying runs in the user's browser



SonicAtlas design (Team 05)

DATASET AND PREPROCESSING DONE

1.2 million tracks cleaned, with every repair measured

1,204,025
tracks in the main audio table

2,777
zero-BPM tempos imputed with MICE

57.5%
lower error than mean imputation (energy, 5% masked)

90,457
re-issues flagged, kept out of trends

24,547
Isolation Forest outliers flagged, not dropped

10.7%
genre table matched to audio table (honest)

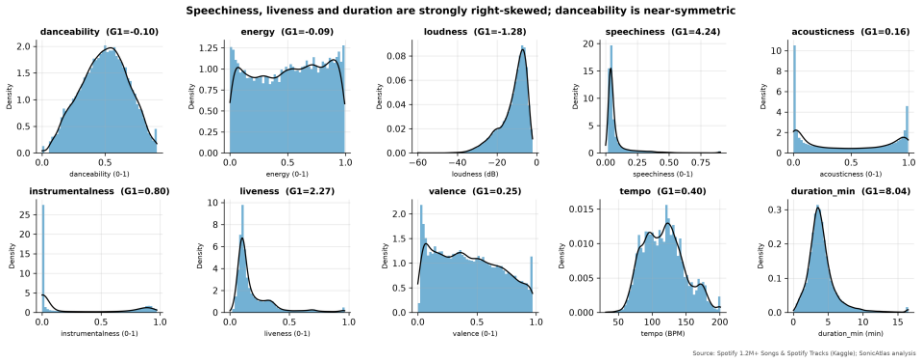
Duration skew G1 8.04 -> 0.23 after log1p; Min-Max and z-score tables stored separately.

Sources: Kaggle rodolfofigueroa/spotify-12m-songs, maharshipandya/-spotify-tracks-dataset, carlosgdcj/genius-song-lyrics-with-language-information

DESCRIPTIVE STATISTICS RESULTS

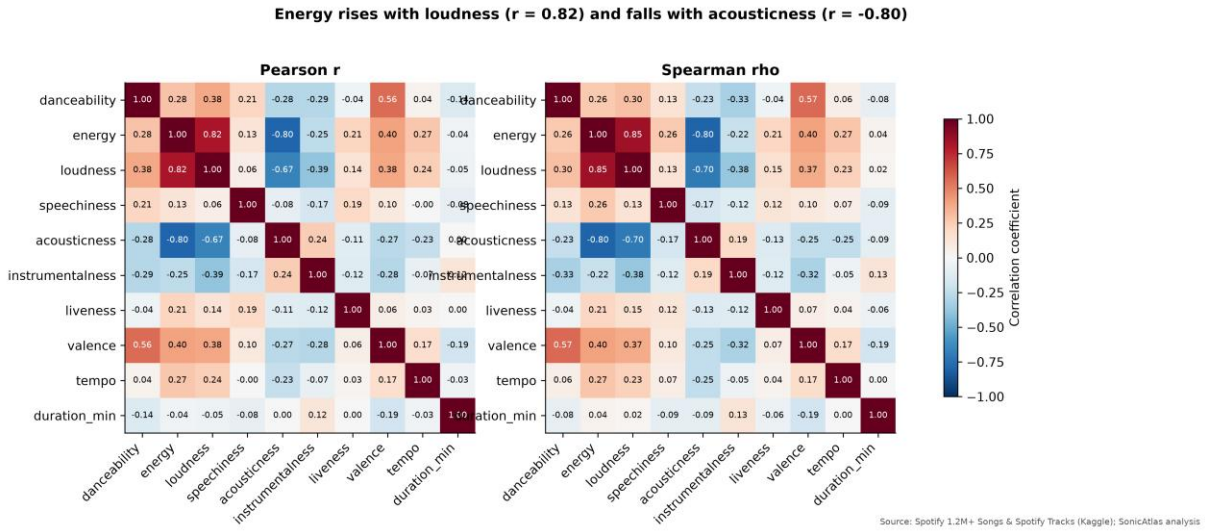
Speechiness and duration are right-skewed; danceability is not

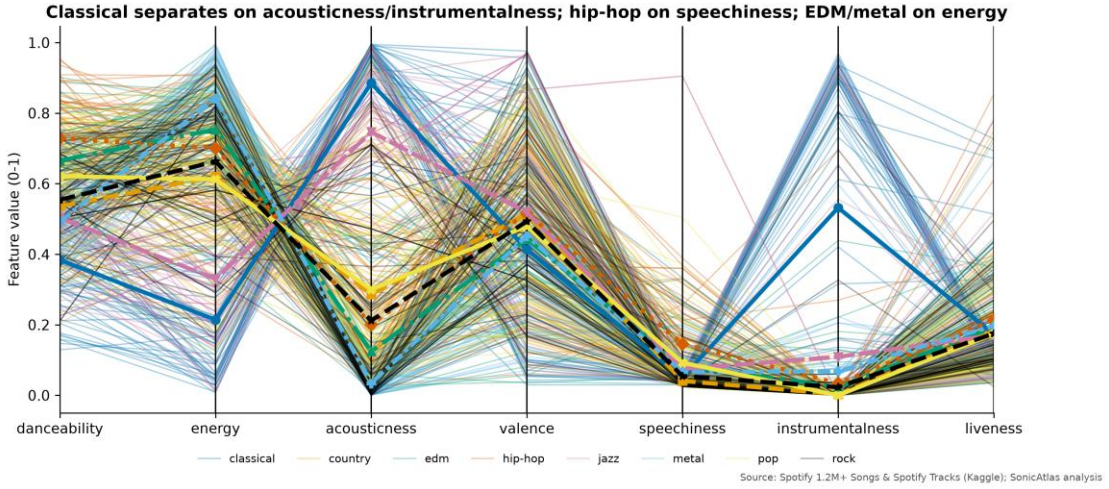
Feature	Mean	Median	Var (n-1)	IQR	G1	G2
danceability	0.493	0.501	0.036	0.277	-0.10	-0.61
energy	0.510	0.524	0.087	0.514	-0.09	-1.22
loudness	-11.809	-9.791	48.750	8.537	-1.28	1.78
speechiness	0.084	0.045	0.013	0.037	4.24	22.35
acousticness	0.447	0.389	0.148	0.823	0.16	-1.64
valence	0.428	0.403	0.073	0.453	0.25	-1.05
tempo	117.899	116.838	925.307	42.753	0.40	-0.29
duration_min	4.147	3.739	7.309	1.863	8.04	146.28



Tukey quartiles (median of halves), n-1 variance, adjusted G1 and bias-corrected G2, unit-tested against SciPy.

Energy follows loudness and opposes acousticness

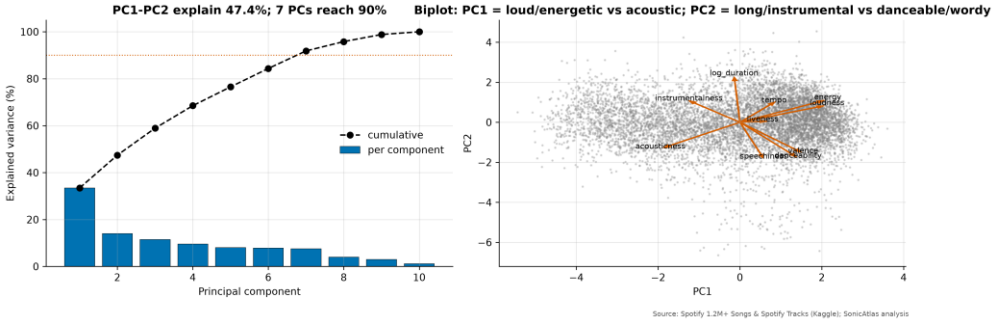


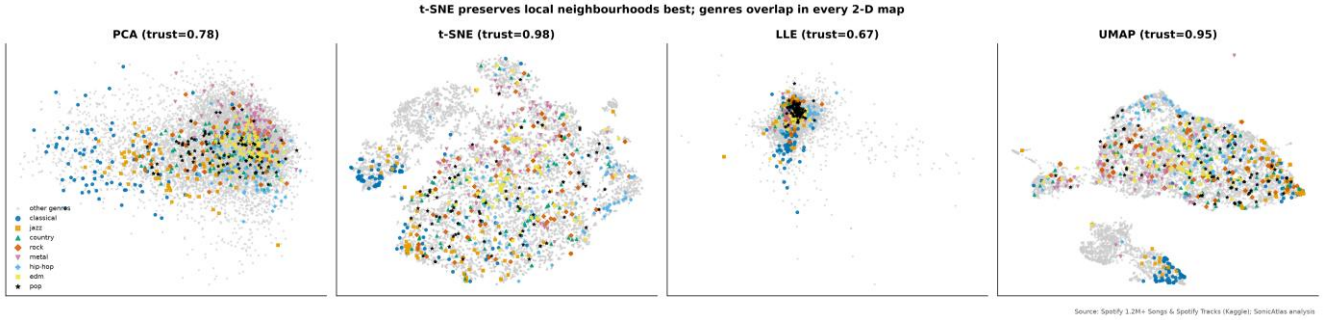


- Loudness ~ energy r = 0.82; energy ~ acousticness r = -0.80
- Valence ~ danceability r = 0.56: happier songs are somewhat more danceable
- Parallel coordinates: classical = acoustic + instrumental; hip-hop = speechy + danceable; metal = highest energy

DIMENSIONALITY REDUCTION FIRST RESULTS

t-SNE keeps local structure best; genres overlap in every map





Method	Runtime (s)	Trustworthiness	Silhouette
PCA	0.0	0.779	-0.085
t-SNE	49.8	0.976	-0.061
LLE	42.4	0.671	-0.140
UMAP	32.2	0.947	-0.055

- PC1 33.5%, PC2 14.0%; 7 PCs reach 90%
- SVD of centred data reproduces PCA exactly
- KMeans k = 6: named clusters such as "Quiet acoustic instrumentals"

MODULE STATUS AND WORK PLAN

Analysis core is complete; the website is the main remaining build



Next: finish the website, lyrics, network and the IEEE paper

Next steps

- Website: Query Studio, Acoustic Map, Compare, Time Machine
- Lyrics sentiment vs audio valence; artist network
- YData-Profiling, Sweetviz, D-Tale benchmark
- Playwright tests, Lighthouse audit, GitHub Pages
- IEEE paper, plagiarism self-check, final report

Key references

[9] J. Serrà et al. (2012), Measuring the Evolution of Contemporary Western Popular Music

[6] M. Interiano et al. (2018), Musical trends and predictability of success in contemporary songs in and out of the top charts

[11] C. Hutto et al. (2014), VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text

[16] L. van der Maaten et al. (2008), Visualizing Data using t-SNE

[25] D. H. da Silva et al. (2025), A Web Application for exploratory data analysis and classification of Parkinson's Disease patients using machine learning models on different datasets

Kaggle datasets: rodolfofigueroa/spotify-12m-songs; maharshipandya/-spotify-tracks-dataset; carlosgdcj/genius-song-lyrics-with-language-information